Assignment 1: Max Warriner (126104460)

1.

   a. See R script
   b. Statistic = 0.3192377
   c. See R script
   d.
      i. Test statistic values: 0.2724547, 0.3024624, 0.2635270
      ii. P-value: 0.017
   e. See R script
   f.
      i. Test statistic values: 0.3068875, 0.2633202, 0.2788457
      ii. P-value: 0.02
   g.

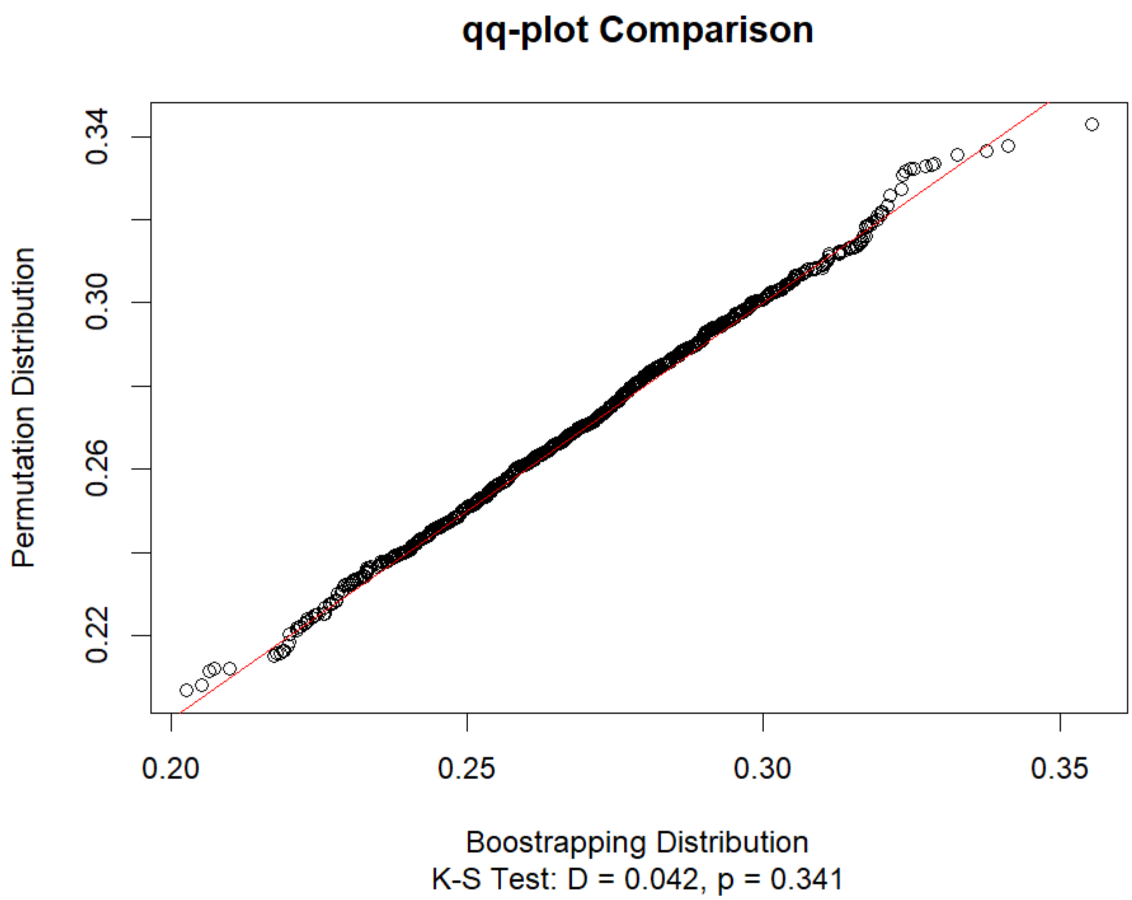


Figure 1: qq-plot comparison of the test statistic null distributions

A qq-plot comparing the boostrapping distribution and the permutation distribution follows closely to a linear y = x line, showing highly similar distributions, apart from some deviance at the tails. A Kolmogorov-Smirnov test of similarity of distributions between the permutation distribution and the boostrapping distribution leads us to fail to reject the null hypothesis that the distributions are the same ($D = 0.042$, $p = 0.341$).

2.

a. Test statistic: 1.885156

b. SE estimate: 0.005111134

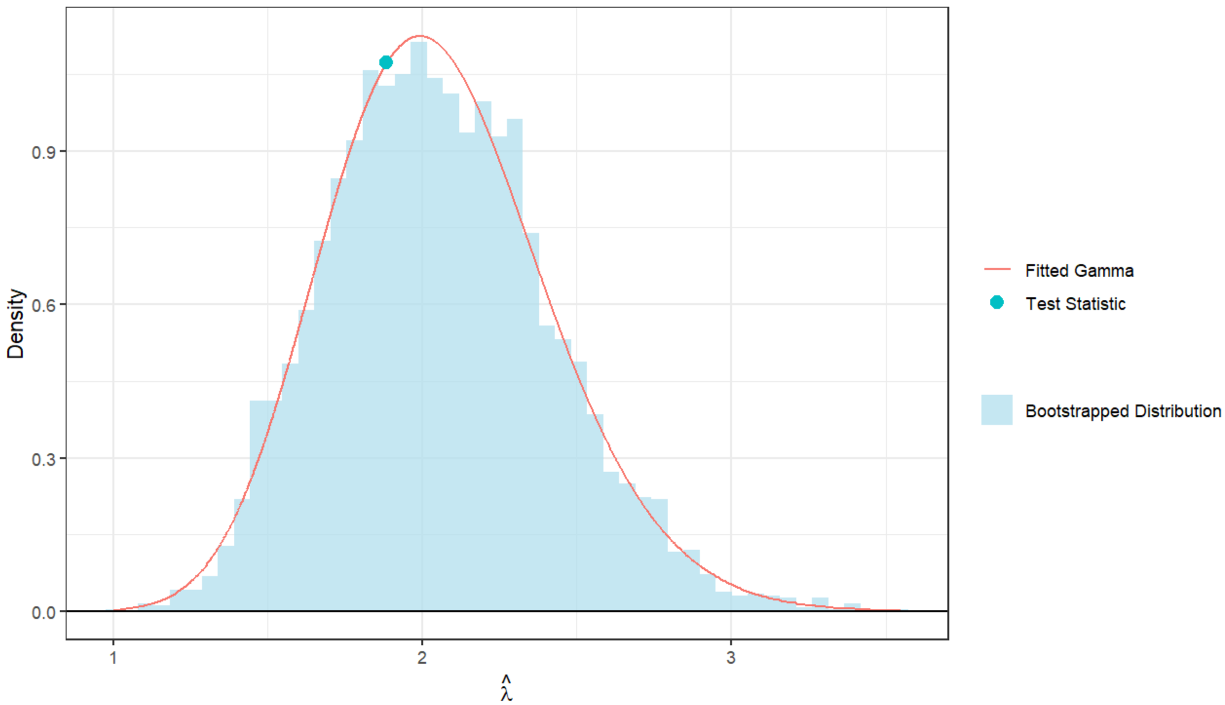


Figure 2: Bootstrapped distribution of the lambda test statistic

c. P-value: 0

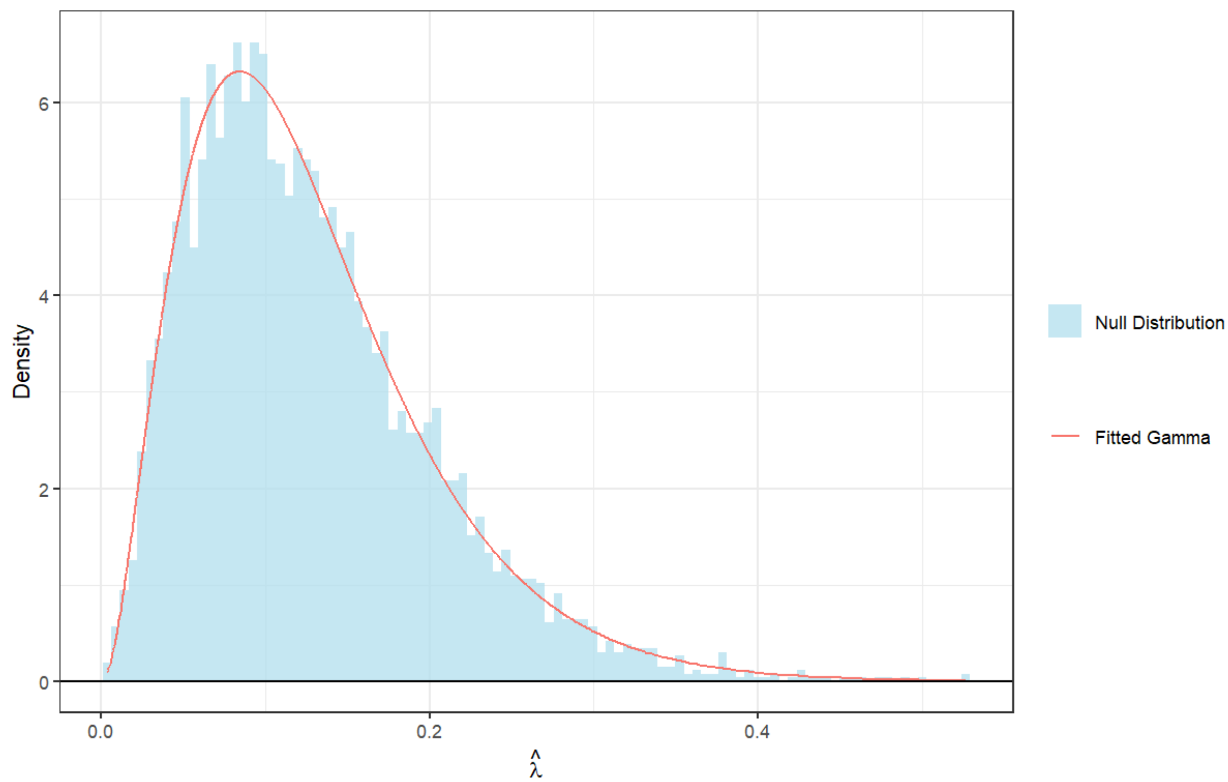


Figure 3: Null distribution of lambda under normality assumption

d. P-value: 0

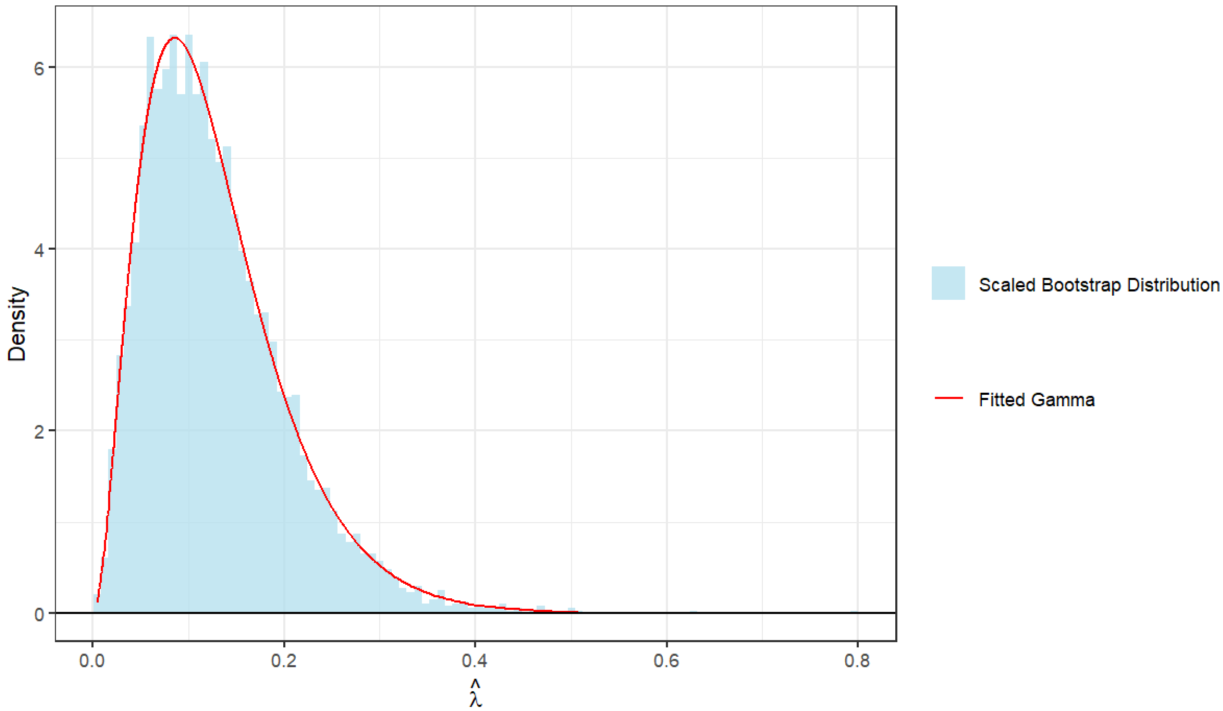


Figure 4: Null distribution of lambda under i.i.d. assumption

e. KS-Test

   i. D = 0.0142

   ii. P-value: 0.6945

3.

a.

3. a. $Cov(X_j, X_k) = \mathbb{E}[(X_j - \mathbb{E}X_j)(X_k - \mathbb{E}X_k)]$, $1 \le j, k \le p$

$$= \mathbb{E}[X_j X_k - X_j \mathbb{E}X_k - X_k \mathbb{E}X_j + \mathbb{E}X_j \mathbb{E}X_k]$$

$$= \mathbb{E}[X_j X_k] - \mathbb{E}[X_j \mathbb{E}X_k] - \mathbb{E}[X_k \mathbb{E}X_j] + \mathbb{E}[\mathbb{E}X_j \mathbb{E}X_k]$$

$$= \mathbb{E}[X_j X_k] - \mathbb{E}X_j \mathbb{E}X_k - \cancel{\mathbb{E}X_k \mathbb{E}X_j} + \cancel{\mathbb{E}X_j \mathbb{E}X_k}$$

$$= \mathbb{E}[X_j X_k] - \mathbb{E}X_j \mathbb{E}X_k \quad //$$

b.

b. $Y = X_1 + X_2$

$$\mathbb{E}[Y] = \mathbb{E}[X_1 + X_2] = \mathbb{E}[X_1] + \mathbb{E}[X_2] \Rightarrow \mu_Y = \mu_{X_1} + \mu_{X_2}$$

$$Cov(Y) = Cov(X_1 + X_2)$$

$$= \mathbb{E}[(X_1 + X_2 - \mathbb{E}[X_1 + X_2])(X_1 + X_2 - \mathbb{E}[X_1 + X_2])^T]$$

$$= \mathbb{E}[(X_1 + X_2 - \mu_1 - \mu_2)(X_1 + X_2 - \mu_1 - \mu_2)^T]$$

$$= \mathbb{E}[((X_1 - \mu_1) + (X_2 - \mu_2))((X_1 - \mu_1) + (X_2 - \mu_2))^T]$$

$$= \mathbb{E}[(X_1 - \mu_1)(X_1 - \mu_1)^T + (X_1 - \mu_1)(X_2 - \mu_2)^T + (X_2 - \mu_2)(X_1 - \mu_1)^T + (X_2 - \mu_2)(X_2 - \mu_2)^T]$$

$$= \mathbb{E}[(X_1 - \mu_1)(X_1 - \mu_1)^T] + \mathbb{E}[(X_1 - \mu_1)(X_2 - \mu_2)^T] + \mathbb{E}[(X_2 - \mu_2)(X_1 - \mu_1)^T] + \mathbb{E}[(X_2 - \mu_2)(X_2 - \mu_2)^T]$$

$$= Var(X_1) + Cov(X_1, X_2) + Cov(X_2, X_1) + Var(X_2) \quad //$$

c.

3.c.

$$\varphi_{X_1}(t) = e^{it^T\mu_1 - \frac{1}{2}t^T\Sigma_1 t}$$

$$\varphi_{X_2}(t) = e^{it^T\mu_2 - \frac{1}{2}t^T\Sigma_2 t}$$

$$\varphi_Y(t) = \mathbb{E}[e^{it^T(X_1+X_2)}] = \mathbb{E}[e^{it^TX_1} e^{it^TX_2}]$$

by X_1 and X_2 independence $\Rightarrow = \mathbb{E}[e^{it^TX_1}]\,\mathbb{E}[e^{it^TX_2}]$

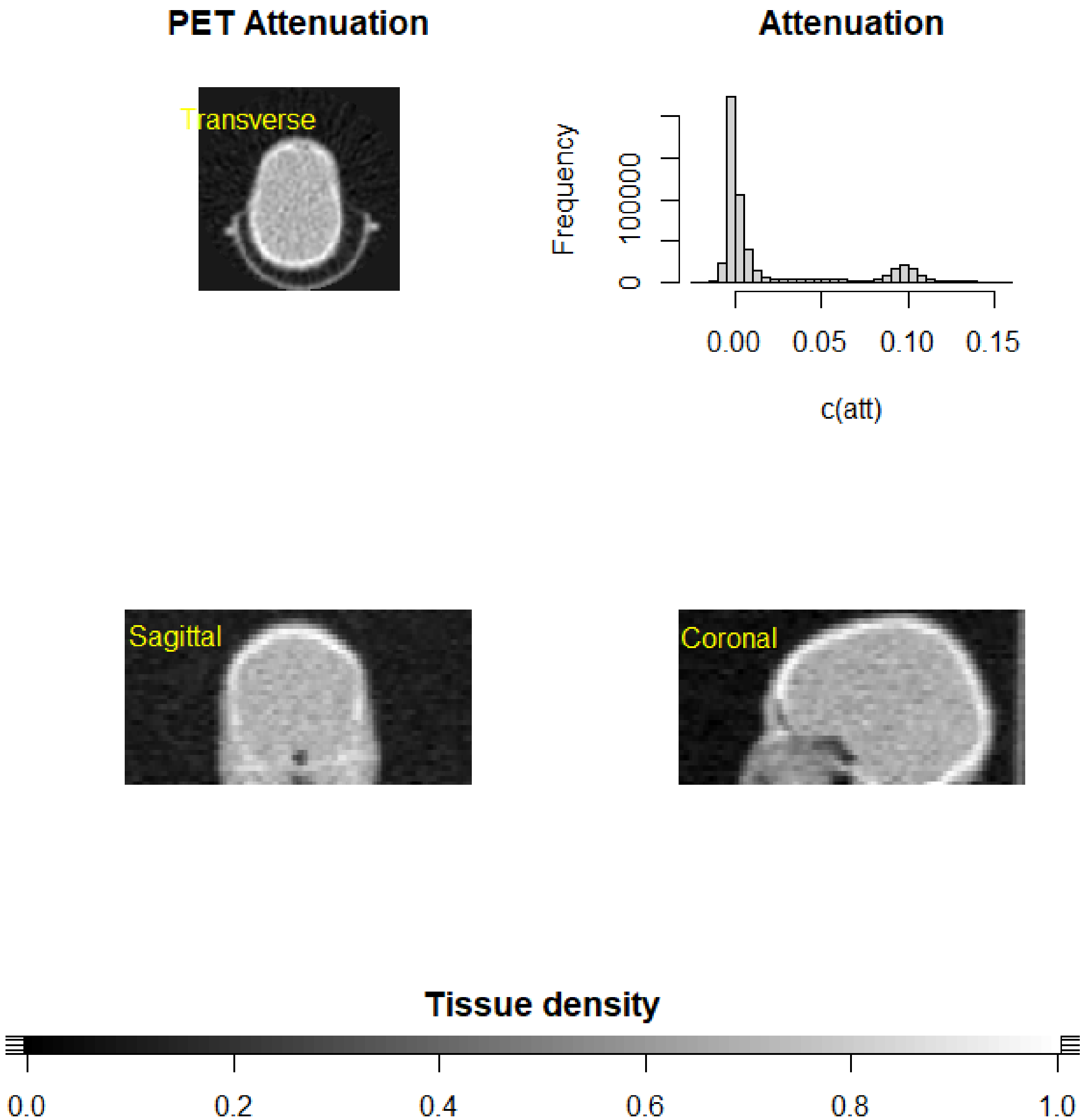
$$= \varphi_{X_1}(t) \cdot \varphi_{X_2}(t)$$

$$= e^{it^T\mu_1 - \frac{1}{2}t^T\Sigma_1 t} \cdot e^{it\mu_2 - \frac{1}{2}t^T\Sigma_2 t}$$

$$= e^{it^T(\mu_1+\mu_2) - \frac{1}{2}t^T(\Sigma_1+\Sigma_2)t}$$

$\therefore Y \sim N_p(\mu_1+\mu_2, \Sigma_1+\Sigma_2)$ //

d.


PET Attenuation
Transverse
Attenuation
Frequency
0
100000
0.00
0.05
0.10
0.15
c(att)
Sagittal
Coronal
Tissue density
0.0
0.2
0.4
0.6
0.8
1.0